Fiscal Year:	FY 2012	**Task Last Updated:**	FY 05/16/2012
PI Name:	Ferl, Robert Ph.D.		
Project Title:	The Impact of Spaceflight on Arabidopsis: Deep Sequencing and DNA Arrays as Collaborative Readouts of the Transcriptome of Arabidopsis Seedlings and Undifferentiated Cells in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	robferl@ufl.edu	**Fax:**	FY 352-392-4072
PI Organization Type:	UNIVERSITY	**Phone:**	352-392-1928
Organization Name:	University of Florida		
PI Address 1:	Horticultural Sciences, 1301 Fifield Hall		
PI Address 2:	Box 110690		
PI Web Page:	http://www.hos.ufl.edu/		
City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences BRIC NNH09ZTT004N
Start Date:	03/01/2010	**End Date:**	02/13/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 NOTE: End date is 2/13/12 per NSSC info; also grant number NNX11AE48G was added (Ed., 4/3/2012) NOTE: Received NCE until 5/31/2012 (original end date was 2/28/2011) per PI (Ed., 2/3/11)		
Key Personnel Changes/Previous PI:	PI Anna-Lisa Paul ; CoPI Robert J. Ferl		
COI Name (Institution):	Paul, Anna-Lisa Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	NNX10AF45G ; NNX11AE48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of our program is to understand fundamental molecular biological responses to extraterrestrial environments. In that context, the goal of this proposal is to compare and contrast the gene expression responses within two forms of Arabidopsis: whole, etiolated seedlings, and undifferentiated cells in culture. Two complementary types of genome-wide gene expression profiling analyses will be applied to the biological samples from both spaceflight and ground controls: Replicated DNA microarray analyses that will reveal changes in genome-wide patterns of gene expression on a platforms whose data are directly comparable to the wide range of array data available for Arabidopsis, and RNA deep sequencing, RNASeq, a relatively new, digital transcript profiling method that offers a more accurate genome-wide quantitation of transcript abundance over wider dynamic range of expression levels. The baseline transcriptome profiling of seedlings and cultured cells addresses the fundamental hypothesis of the proposal - that spaceflight is a unique and challenging environment in which survival requires adaptive changes that are both governed and displayed by alterations in gene expression. The comparison of intact plants with cultures of undifferentiated cells addresses a second hypothesis that cells can detect spaceflight and gravity in the absence of tissue or organized developmental structures. These experiments enable us to evaluate fundamental mechanisms associated with plant responses to the novel environment of spaceflight. What gene response pathways are engaged in response to spaceflight? Are specific organs and tissues required for the detection and transduction of signals from the spaceflight environment, including gravity? Given that certain differentiated cells are often associated with the perception or signaling of environmental events, what happens when they are absent? Would gravity still exert an effect on plant cells? Can single cells perceive gravity? These key questions are recognized as fundamental within the previous decadal study and underpin some of the biggest questions in spaceflight plant biology. And while several studies have addressed gene expression in space, as yet there is no robust, replicated transcriptome analysis of Arabidopsis in spaceflight. This project contains the replicates and technical sophistication to form those baseline data for seedlings and extends those data into essential questions regarding gravity and spaceflight perception by developed tissues and organs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These experiments enable us to evaluate fundamental mechanisms associated with plant responses to the novel environment of spaceflight. What gene response pathways are engaged in response to spaceflight? Are specific organs and tissues required for the detection and transduction of signals from the spaceflight environment, including gravity? Given that certain differentiated cells are often associated with the perception or signaling of environmental events, what happens when they are absent? Would gravity still exert an effect on plant cells? Can single cells perceive gravity? These key questions are recognized as fundamental within the previous decadal study and underpin some of the biggest questions in spaceflight plant biology.
	Arabidopsis seedlings and cell cultures clearly respond to the spaceflight environment with changes in gene expression. Microarray, SAGE, and quantitative PCR data fully support the conclusion that Arabidopsis, whether as etiolated seedlings or undifferentiated cells, senses that it is in a modified environment and mounts changes in gene expression as a response to spaceflight. The six-fold replication of the experiment design allowed an examination of differences in gene expression down to two-fold while retaining p values <0.01, allowing a confident examination of fairly small quantitative changes in gene expression At a minimum, any genes affected by gravity or orientation on Earth might be expected to be changed in spaceflight. However, the range of genes affected by spaceflight appears to span beyond those genes whose expression might be easily explained by changes in gravity, and include a number of familiar genes with roles in the response to a wide variety of environmental factors. In addition, the results of these experiments reflect relatively long term adaptive responses as the plants and cells were on orbit for twelve days before they were fixed. Thus the patterns of gene expression reflect adaptive strategies of these two forms of the plant, rather than a rapid initial response to a new stress or environment. Time course analyses of any number of stress response surveys repeatedly show that initial responses are often diminished within minutes of the application of the stress, and patterns of gene expression change as longer term adaptive strategies are engaged and maintained. Thus these spaceflight-induced changes in gene expression likely reflect changes toward longer-term survival in the spaceflight environment. Future experiments should include the initial responses to spaceflight. In etiolated seedlings, the genes most highly induced by spaceflight (by five-fold or more) are almost equally divided between those associated with pathogen defense and those that are associated with environmental stress responses. Differentially regulated genes reflect categories of genes which are broadly associated with drought stress, wounding and calcium- and auxin-mediated signaling. The seedlings grew on sterile media and none experienced drought or wounding stress. The temperature was well controlled and any excursions from the Ground Control environmental profiles were minimal. The media compositions and atmospheric conditions were identical between Spaceflight and Ground Control. Therefore a direct and obvious relationship between the affected genes and a defined stress is missing. A possible basis for many of these responses may be traced to a translation of changes in forces associated with the cytoskeleton and cell wall. Pathogen and wounding responses are very tightly connected, as both can involve a breach of the cell wall. In the Tensegrity-based model of the gravisensing, an actin-based cytoskeletal network throughout the cytoplasm is coupled to stretch-sensitive receptors in the plasma membrane (Yoder et al., 2001). Cytoskeletal actin filaments attached to the plasma membrane, specialized proteins that contain an actin binding domain at their carboxyl terminus, and protein phosphorylation or phosphoinositide binding (Tsukita and Yonemura, 1997) have been postulated to participate in gravity related signaling (Yoder et al., 2001). It is possible that any of these signaling mechanisms by which a cell detects a breach are balanced by the force of gravity and the absence of gravity may confound the signaling system, engaging pathogen signal transduction pathways as a consequence. The presence of other stress response genes may have similar explanations; they appear inappropriate only because the foundations of the stimulus are unclear. About 20% of the up-regulated genes in seedlings do not fall obviously into either pathogen or environmental stresses. It is from among these examples that it may be possible to identify some “appropriate”, or at least adaptively meaningful, responses to the spaceflight environment. Several candidates in this category include ALF4 (Aberrant Lateral Root Formation 4), ARR7 (Response Regulator 7), and PIN3 (Pin-Formed 3), all of which are either influenced by, or play a role in auxin or cytokinin signaling, and may be impacted by changes in the gravity vector (Lomax, 1997; Friml et al.,

Task Progress:

2002; DiDonato et al., 2004; Perrin et al., 2005; Aloni et al., 2006; Harrison and Masson, 2008a).

The cells in culture responded in a dramatically different fashion to spaceflight than did the seedlings. While similar numbers of genes were differentially affected by spaceflight, there is almost no similarity among the identities of the genes affected. In addition, the cell cultures exhibited a greater number of highly (greater than 8-fold) differentially expressed genes. In seedlings, most of the genes differentially expressed by two-fold or more had low intrinsic signal values, while nearly all of the differentially expressed genes in cultured cells had moderate to high signal values. About one-third of the differentially expressed cell culture genes had expression levels of greater than 2000, while only four of the differentially expressed genes from seedlings were as abundant. These data suggest that cell cultures mount a more dramatic response to spaceflight, affecting more genes with higher expression levels and with greater fold change.

None of the genes significantly impacted by spaceflight in cultured cells were similar to those in seedlings. Even the GO categories of genes are substantially different. Cell cultures also exhibit less diversity in the types of genes which are highly induced. Pathogen and wounding, general stress response and transcription factor categories are present, as in the seedling data sets, but the genes that populate those categories differ from those induced in seedlings. The cell culture samples illustrate repression of more kinases, and fewer proteases than seedlings, but transcription factors, transporters, and cell wall associated gene categories are about equally affected – however, with different individual genes in each case.

By far the most dramatic molecular response to spaceflight in the cultured cells was a heat shock response, a response that was not manifested in seedlings. Genes encoding heat shock proteins, heat shock related proteins or heat shock transcription factors, comprise the majority of the genes induced greater than 8-fold and over 40% of the those induced more than five-fold. Clearly no temperature differentials capable of classic heat shock were present between the Spaceflight samples and the Ground Control samples. Heat shock proteins are, however, generally related to responses to unfolded proteins and can be part of a variety of signal transduction pathways. It has been shown in both plants and animals that heat shock-related proteins are widely distributed in response to an assortment of abiotic stresses. In addition to their roles as molecular chaperones that manage post-translational processes, heat shock proteins are thought to participate in the cross talk among stress response pathways, especially where an organism is exposed to multiple, concomitant stresses. Over-expression of heat shock proteins may contribute to a generalized tolerance for multiple environmental stresses (Swindell et al., 2007). In a situation involving a novel environment such as spaceflight, it is possible that a plant cell responds as if it were being exposed to multiple stressors, and the induction of heat shock proteins reflects this perception.

There are few previous studies to provide ground for direct comparison with the current data, yet interesting correlations emerge. Three genome-scale plant molecular space studies are published to date (Paul et al., 2005; Stutte et al., 2006; Salmi and Roux, 2008). One positive correlation is that evidence of spaceflight heat shock was seen in Arabidopsis (Paul and others 2005), albeit in light-grown seedlings quite distinct from those of the current study. Another positive, but less direct, correlation is the effect of spaceflight on the expression of genes involved in calcium signaling during fern spore development (Salmi and Roux 2008). However, there were virtually no statistically supportable changes in the patterns of gene expression between wheat for comparison (Stutte and others 2006). Genome-wide studies of the impact of spaceflight on animal systems are slightly more common, particularly for cell lines from bone-derived cultures, mice and drosophila models. While a full review of that literature is beyond the scope of this study, recent studies do demonstrate clear transcriptome differences during spaceflight, including studies of murine muscle (Allen et al., 2009) and thymus tissue (Lebsack et al., 2010). Drosophila and Caenorabditis elegans show six differentially expressed genes common to these two diverse species (Leandro et al., 2007; Adenle et al., 2009). And while the divergence of plants and animals largely prevents specific gene comparisons in spaceflight responses, certain classes of genes seem similarly affected by spaceflight. Calcium mediated signal transduction is impacted by spaceflight in both plants and animals (Hughes-Fulford et al., 1998; Hashemi et al., 1999; Paul et al., 2001; Salmi and Roux, 2008) and genes affected by calcium signaling are differentially expressed. Heat shock proteins are highly conserved in both kingdoms and heat shock genes are upregulated in some spaceflight studies. At least one of the sub-class of heat shock genes highly induced in this study (Hsp90) is also induced in mouse thymus after exposure to spaceflight (Lebsack et al., 2010). Stress also exacerbates sensitivity to pathogen attach in both plants and animals and there are common mechanisms for triggering an immune response, such as nitric oxide and mitogen-activated protein kinase cascades (Nurnberger et al., 2004). A number of genes associated with Drosophila immune pathways were downregulated in the spaceflight environment, and larvae were in fact more susceptible to pathogen infection (Marcu et al., 2011). Although none of the Drosophila genes in that study are orthologous to the Arabidopsis genes presented here, many of genes in the present seedling data are nonetheless associated with pathogen defense. While many more studies will be required to support informed deductions regarding commonalities in spaceflight response (especially considering the dissimilarity of biology, hardware, flight profiles, and even replicates), current data suggest that such commonalities may exist.

That plants and plant cells under BRIC experimental conditions respond to spaceflight with significant changes in gene expression indicate that changes in metabolism are engaged to cope with spaceflight. Spaceflight demands an adaptive response. Many of these changes make some sense with regard to the primary effects of gravity, but others suggest stresses that are not simply explained by the lack of gravity. That cultured cells respond dramatically to spaceflight in a markedly different way than seedlings, suggests that either undifferentiated cells have more biological difficulty with spaceflight or that there are technical reasons affecting the results. One possibility is that this difference between cultured cells and seedlings is due to the fact seedlings have organs with which to sense and sample their environment, and that the absence of such organs undifferentiated cells propagate inappropriate stress responses. In support of this possibility, seedlings displayed a greater abundance of differentially expressed genes associated with plant hormone signaling and gravity sensing while cultured cells did not. Another possible explanation for the differences is simply that all of the cells in the culture are responding in an identical manner, enhancing the technical observation of the response, while in seedlings different tissues are responding differently, diluting the apparent effect as induction in one tissue balances repression in another. The analysis of spaceflight effects on seedlings would benefit from recovering enough material to look at individual organs or cell types. Regardless, it remains clear that there is not an intense and simple response to spaceflight that is shared by all cell types within the differentiated cells of an intact plant. Conclusions

Seedlings and cell cultures both engage gene expression changes to alter metabolic responses to the spaceflight environment, but they engage dramatically different responses. This response differential between seedlings and cell cultures could be as simple as a consequence of examining a very uniformly responding system in undifferentiated cells compared to a complex organism composed of differentiated organs and tissues. Or, it could be as complex of the presence of an organ-specific feedback sense and response system that allows seedlings, but not undifferentiated cells, to more properly sense the environment and integrate a response that makes sense based on input from organ systems.

Regardless, it clear that even undifferentiated plant cells sense spaceflight as an environment that demands molecular responses. Many of those responses tap familiar pathways - auxin, calcium signaling, stress – and there is a tendency to place those responses into familiar earth-based context. But given the truly novel nature of spaceflight, responses may propagate in ways that are distinct from well-defined terrestrial analogs.

Additional details can be found in publication: Paul A-L, Zupanska A, Ostrow DT, Zhang Y, Sun Y, Li J-L, Shanker S, Farmerie WG, Amalfitano CE, Ferl RJ (2012) Spaceflight transcriptomes: unique responses to a novel environment. Astrobiology 12: 40-56.

Literature Cited:

Adenle AA, Johnsen B, Szewczyk NJ (2009) Review of the results from the International C. elegans first experiment (ICE-FIRST). Adv Space Res 44: 210-216

Allen DL, Bandstra ER, Harrison BC, Thorng S, Stodieck LS, Kostenuik PJ, Morony S, Lacey DL, Hammond TG, Leinwand LL, Argraves WS, Bateman TA, Barth JL (2009) Effects of spaceflight on murine skeletal muscle gene expression. J Appl Physiol 106: 582-595

Aloni R, Aloni E, Langhans M, Ullrich CI (2006) Role of cytokinin and auxin in shaping root architecture: regulating vascular differentiation, lateral root initiation, root apical dominance and root gravitropism. Ann Bot (Lond) 97: 883-893

DiDonato RJ, Arbuckle E, Buker S, Sheets J, Tobar J, Totong R, Grisafi P, Fink GR, Celenza JL (2004) Arabidopsis ALF4 encodes a nuclear-localized protein required for lateral root formation. The Plant journal : for cell and molecular biology 37: 340-353

Friml J, Wisniewska J, Benkova E, Mendgen K, Palme K (2002) Lateral relocation of auxin efflux regulator PIN3 mediates tropism in Arabidopsis. Nature 415: 806-809

Harrison BR, Masson PH (2008a) ARL2, ARG1 and PIN3 define a gravity signal transduction pathway in root statocytes. Plant J 53: 380-392

Hashemi BB, Penkala JE, Vens C, Huls H, Cubbage M, Sams CF (1999) T cell activation responses are differentially regulated during clinorotation and in spaceflight. Faseb J 13: 2071-2082

Hughes-Fulford M, Tjandrawinata R, Fitzgerald J, Gasuad K, Gilbertson V (1998) Effects of microgravity on osteoblast growth. Gravit Space Biol Bull 11: 51-60

Leandro LJ, Szewczyk NJ, Benguria A, Herranz R, Lavan D, Medina FJ, Gasset G, Loon JV, Conley CA, Marco R (2007) Comparative analysis of Drosophila melanogaster and Caenorhabditis elegans gene expression experiments in the European Soyuz flights to the International Space Station. Adv Space Res 40: 506-512

Lebsack TW, Fa V, Woods CC, Gruener R, Manziello AM, Pecaut MJ, Gridley DS, Stodieck LS, Ferguson VL, Deluca D (2010) Microarray analysis of spaceflown murine thymus tissue reveals changes in gene expression regulating stress and glucocorticoid receptors. J Cell Biochem 110: 372-381

Lomax TL (1997) Molecular genetic analysis of plant gravitropism. Gravit Space Biol Bull 10: 75-82

Marcu O, Lera MP, Sanchez ME, Levic E, Higgins LA, Shmygelska A, Fahlen TF, Nichol H, Bhattacharya S (2011) Innate immune responses of Drosophila melanogaster are altered by spaceflight. PLoS ONE 6: e15361

Nurnberger T, Brunner F, Kemmerling B, Piater L (2004) Innate immunity in plants and animals: striking similarities and obvious differences. Immunol Rev 198: 249-266

Paul AL, Daugherty CJ, Bihn EA, Chapman DK, Norwood KL, Ferl RJ (2001) Transgene expression patterns indicate that spaceflight affects stress signal perception and transduction in arabidopsis. Plant Physiol 126: 613-621

Paul AL, Popp MP, Gurley WB, Guy C, Norwood KL, Ferl RJ (2005) Arabidopsis gene expression patterns are altered during spaceflight. Adv Space Res 36: 1175-1181

Perrin RM, Young LS, Murthy UMN, Harrison BR, Wang Y, Will JL, Masson PH (2005) Gravity signal transduction in primary roots. Ann Bot (Lond) 96: 737-743

Salmi ML, Roux SJ (2008) Gene expression changes induced by space flight in single-cells of the fern Ceratopteris richardii. Planta 229: 151-159

Stutte GW, Monje O, Hatfield RD, Paul AL, Ferl RJ, Simone CG (2006) Microgravity effects on leaf morphology, cell structure, carbon metabolism and mRNA expression of dwarf wheat. Planta 224: 1038-1049

Swindell WR, Huebner M, Weber AP (2007) Transcriptional profiling of Arabidopsis heat shock proteins and transcription factors reveals extensive overlap between heat and non-heat stress response pathways. BMC Genomics 8: 125

Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Paul A-L, Zupanska A, Ostrow DT, Zhang Y, Sun Y, Li J-L, Shanker S, Farmerie WG, Amalfitano CE, Ferl RJ. "Spaceflight transcriptomes: unique responses to a novel environment." Astrobiology. 2012 Jan;12(1):40-56. PubMed PMID: 22221117 ; http://dx.doi.org/10.1089/ast.2011.0696 , Jan-2012